

Quantum Mechanics

Lecture 4: Continuum States, Quantized Momentum, and Interference

Lectures by Leonard Susskind

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Chapter 1

Continuum States, Quantized Momentum, and Interference

1.1 When a Discrete Label Becomes Continuous

Begin with an observable whose possible values are labeled by integers $n = 1, 2, \dots$. Its probabilities obey

$$\sum_n P(n) = 1, \quad (1.1)$$

and the corresponding basis vectors may be chosen orthonormally:

$$\langle m | n \rangle = \delta_{mn}. \quad (1.2)$$

The Kronecker delta is one when its two labels agree and zero otherwise. It can equally well be regarded as a function of the difference $m - n$.

For a state $|\psi\rangle$, the probability of obtaining n is

$$P(n) = |\langle n | \psi \rangle|^2. \quad (1.3)$$

These three statements are the pattern we must preserve when the label is no longer discrete.

Let the measured quantity now be the position x of a particle on a line. For an ordinary continuous distribution, a single point has zero probability. The useful object is therefore a probability density $\rho(x)$. The probability of finding the particle in a short interval is

$$\text{Prob}(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} \rho(x) dx, \quad \int_{-\infty}^{\infty} \rho(x) dx = 1. \quad (1.4)$$

Thus sums become integrals, and probability masses become densities.

1.2 The Dirac Delta

The continuum analogue of the Kronecker delta is the Dirac delta distribution. Its informal picture is an infinitely narrow spike at $x = y$, zero away from that point, with unit area:

$$\delta(x - y) = 0 \quad (x \neq y), \quad \int_{-\infty}^{\infty} \delta(x - y) dx = 1. \quad (1.5)$$

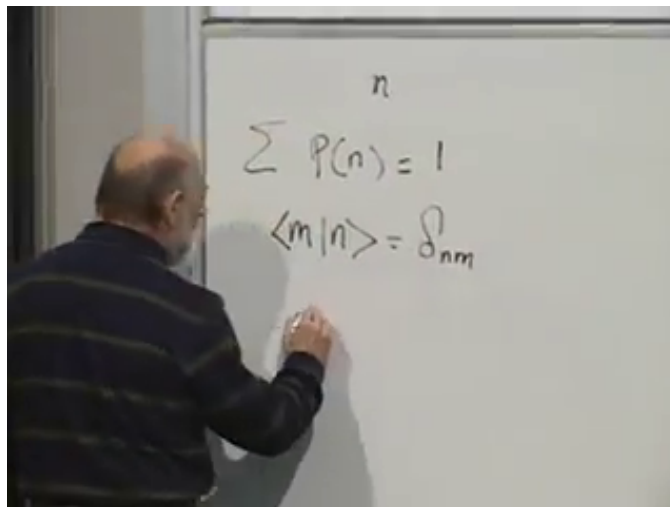


Figure 1.1: Discrete probability normalization and basis orthonormality.

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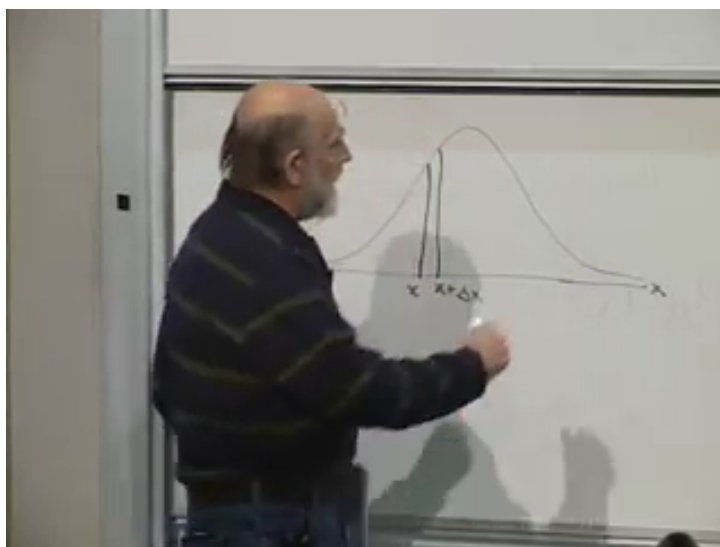


Figure 1.2: For a continuous coordinate, probability belongs to an interval and is represented by area under a density.

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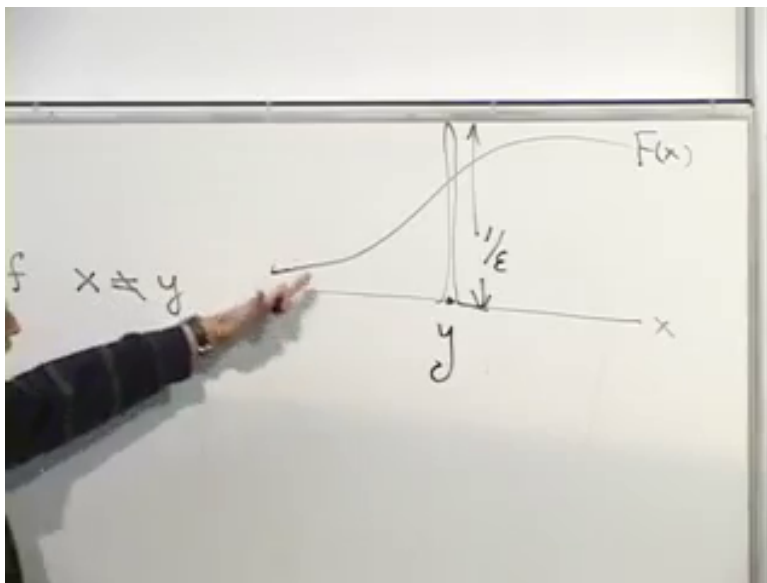


Figure 1.3: A narrow unit-area spike samples the value of a smooth function at $x = y$.

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No ordinary function has these properties. The precise definition is its action on a sufficiently regular test function:

$$\int_{-\infty}^{\infty} \delta(x - y) f(x) dx = f(y). \quad (1.6)$$

The delta is therefore best understood as a distribution under an integral, not as a function with a literal infinite value.

A useful approximation makes the spike picture explicit. Let $\delta_\epsilon(x - y)$ be a rectangle of width ϵ and height $1/\epsilon$, centered at y . Its area is one, and for continuous f ,

$$\int \delta_\epsilon(x - y) f(x) dx \longrightarrow f(y) \quad (\epsilon \rightarrow 0). \quad (1.7)$$

The unit area must not be confused with a unit Hilbert-space norm: $\int |\delta_\epsilon|^2 dx = 1/\epsilon$ diverges in the limit.

Position basis states are consequently delta-normalized:

$$\langle x | y \rangle = \delta(x - y), \quad \int_{-\infty}^{\infty} dx |x\rangle \langle x| = \mathbf{1}. \quad (1.8)$$

Distinct positions are orthogonal. The formal quantity $\langle x | x \rangle = \delta(0)$ is divergent because $|x\rangle$ is a generalized eigenstate, not a normalizable vector in the Hilbert space.

1.3 The Position Amplitude and Born Density

The abstract probability rule has not changed. In the discrete case, $|\langle n | \psi \rangle|^2$ is the probability for the outcome n . In the continuous case,

$$\rho(y) = |\langle y | \psi \rangle|^2 \quad (1.9)$$

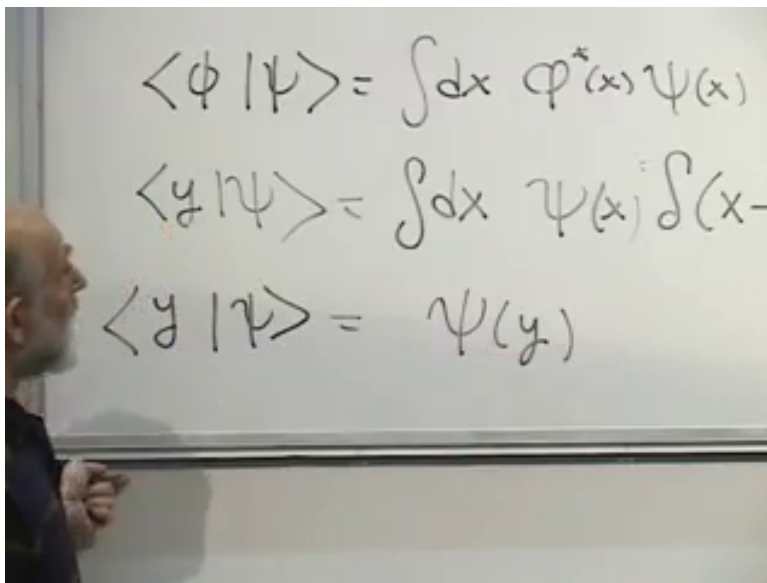


Figure 1.4: Delta sampling identifies the position amplitude $\langle y | \psi \rangle$ with the wavefunction value $\psi(y)$.

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is the probability density at y . To see what this means in the position representation, write

$$\psi(x) := \langle x | \psi \rangle. \quad (1.10)$$

The wavefunction of the generalized position state $|y\rangle$ is $\langle x | y \rangle = \delta(x - y)$. Hence

$$\langle y | \psi \rangle = \int dx \langle y | x \rangle \langle x | \psi \rangle \quad (1.11)$$

$$= \int dx \delta(x - y) \psi(x) = \psi(y). \quad (1.12)$$

Born's rule is therefore

$$\rho(y) = |\psi(y)|^2, \quad \text{Prob}(y \in [a, b]) = \int_a^b |\psi(y)|^2 dy. \quad (1.13)$$

Question from the audience. Could the first probability postulate be stated once more for the continuous case?^a

Response. For a discrete observable, the probability of outcome n is $|\langle n | \psi \rangle|^2$. For position, the same expression $|\langle y | \psi \rangle|^2$ is a density. Since $\langle y | \psi \rangle = \psi(y)$, the probability of an interval is the integral of $|\psi(y)|^2$ over that interval. Dirac's notation was designed so that the discrete and continuous statements have the same form.

^aLecture timestamp: 00:22:39–00:26:37.

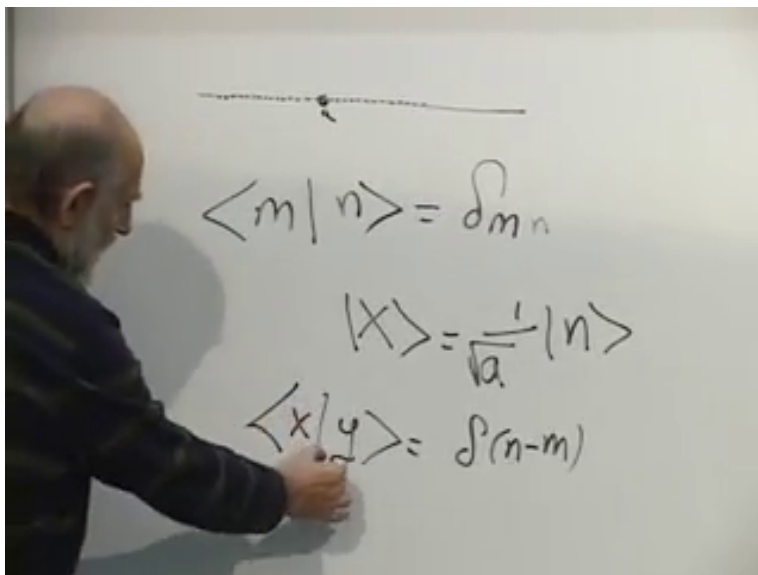


Figure 1.5: Rescaling lattice kets by $a^{-1/2}$ produces Dirac-delta normalization in the continuum limit.

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1.4 A Lattice Explains Continuum Normalization

The divergent self-overlap of $|x\rangle$ becomes transparent if the line is temporarily replaced by a lattice with spacing a . Let $x_n = na$, and let $|n\rangle$ be ordinary normalized site states:

$$\langle n | m \rangle = \delta_{nm}, \quad \sum_n |n\rangle \langle n| = \mathbf{1}. \quad (1.14)$$

Define continuum-normalized states at the lattice points by

$$|x_n\rangle := \frac{1}{\sqrt{a}} |n\rangle. \quad (1.15)$$

Their overlap and completeness relations are

$$\langle x_n | x_m \rangle = \frac{\delta_{nm}}{a}, \quad \sum_n a |x_n\rangle \langle x_n| = \mathbf{1}. \quad (1.16)$$

As $a \rightarrow 0$, the sum $\sum_n a$ becomes an integral, while δ_{nm}/a becomes $\delta(x - y)$. The height $1/a$ and width a again give unit area.

Question from the audience. How can the overlap of a state with itself be infinite without ruining the probability interpretation? ^a

Response. The position eigenstates are not physical normalized states; they are generalized basis vectors. On a lattice their normalized ancestors are perfectly ordinary. The factor $a^{-1/2}$ converts a discrete probability into a density, and it converts δ_{nm} into δ_{nm}/a . Finite wave packets remain normalized, and every delta appears under an integral where its unit-area property gives a finite answer.

^aLecture timestamp: 00:26:38–00:34:24.

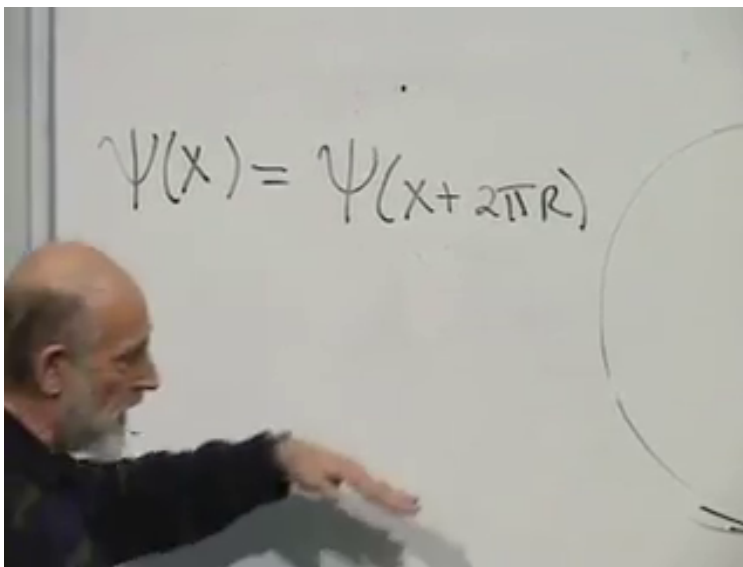


Figure 1.6: Cutting the circle gives an interval whose wavefunction values must match after one circumference.

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1.5 A Particle on a Circle

Now constrain the particle to a circle of radius R . Cutting the circle at one point turns it into the interval

$$0 \leq x < 2\pi R, \quad (1.17)$$

but the two ends represent the same physical point. The wavefunction must therefore satisfy

$$\psi(x + 2\pi R) = \psi(x). \quad (1.18)$$

Periodic functions form a complex vector space: scalar multiples and sums of periodic functions are still periodic.

On this space the momentum operator is

$$\hat{p} = -i\hbar \frac{d}{dx}, \quad (1.19)$$

with periodic boundary conditions included in its domain. Its eigenvalue equation,

$$-i\hbar \frac{d\psi_p}{dx} = p\psi_p, \quad (1.20)$$

has the plane-wave solution

$$\psi_p(x) = Ce^{ipx/\hbar}. \quad (1.21)$$

The name momentum will acquire its full physical justification only after time evolution and the motion of wave packets are developed. For now it is the eigenvalue of the translation generator.

Normalization on the finite circle fixes C :

$$1 = \int_0^{2\pi R} |\psi_p(x)|^2 dx = 2\pi R |C|^2, \quad (1.22)$$

$$\psi_p(x) = \frac{1}{\sqrt{2\pi R}} e^{ipx/\hbar}. \quad (1.23)$$

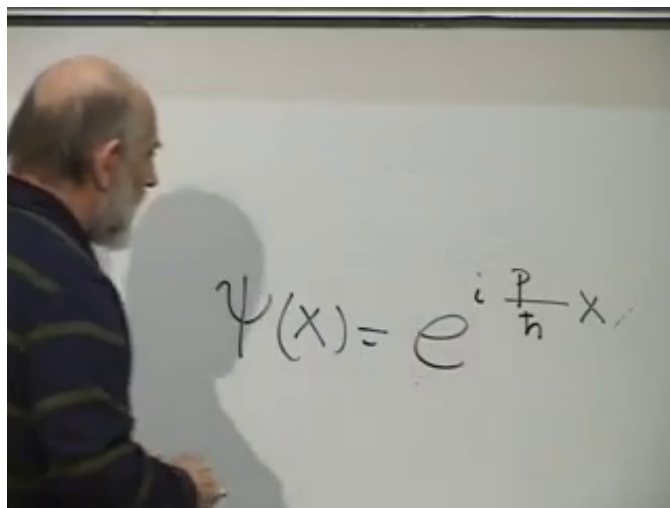


Figure 1.7: The momentum eigenfunction is a complex plane wave.

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The constant magnitude means that a momentum eigenstate has uniform position density around the circle.

1.6 Periodicity Quantizes Momentum

The normalized plane wave is not automatically an allowed state. It must also obey the periodic boundary condition:

$$e^{ip(x+2\pi R)/\hbar} = e^{ipx/\hbar}, \quad (1.24)$$

$$e^{i2\pi Rp/\hbar} = 1. \quad (1.25)$$

The phase gained in one circuit must be an integer multiple of 2π :

$$\frac{2\pi Rp}{\hbar} = 2\pi n, \quad n \in \mathbb{Z}. \quad (1.26)$$

Thus

$$p_n = \frac{n\hbar}{R}, \quad \psi_n(x) = \frac{1}{\sqrt{2\pi R}} e^{inx/R}. \quad (1.27)$$

The allowed functions are mutually orthogonal:

$$\int_0^{2\pi R} \psi_m^*(x) \psi_n(x) dx = \delta_{mn}. \quad (1.28)$$

Here position is continuous while momentum is discrete. In this problem the discreteness is a consequence of compactness and periodicity; discrete spectra in other systems can arise from other boundary conditions or from a confining potential.

The spacing between neighboring momenta is

$$\Delta p = \frac{\hbar}{R}. \quad (1.29)$$

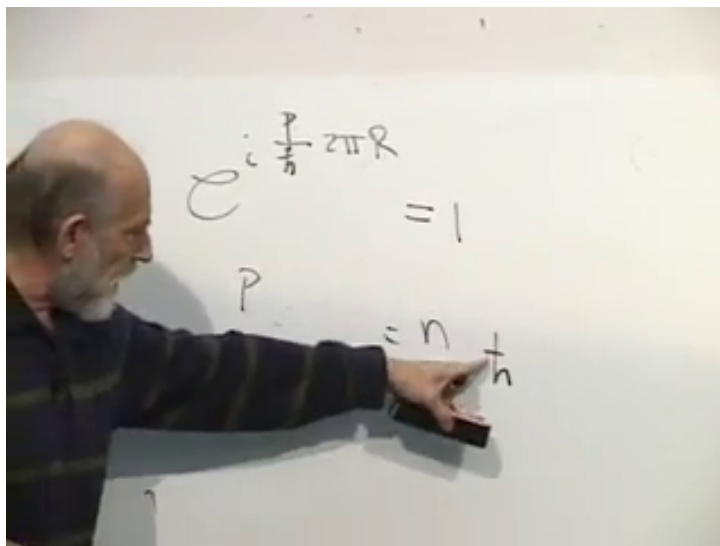


Figure 1.8: Single-valuedness around the circle restricts momentum to $p_n = n\hbar/R$.

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As $R \rightarrow \infty$, this spacing tends to zero. A continuum-normalized momentum basis may be obtained by the same rescaling used for position:

$$|p_n\rangle_c := \frac{1}{\sqrt{\Delta p}} |n\rangle, \quad {}_c\langle p_m | p_n \rangle_c = \frac{\delta_{mn}}{\Delta p} \rightarrow \delta(p - p'). \quad (1.30)$$

For orbital motion on the circle, the angular momentum is

$$L = Rp, \quad (1.31)$$

and therefore

$$L_n = Rp_n = n\hbar. \quad (1.32)$$

The radius cancels because angular momentum measures phase winding around an angle. This conclusion applies to a scalar orbital wavefunction with $\psi(\phi + 2\pi) = \psi(\phi)$. Half-integer spin belongs to a different representation and is a later subject.

Question from the audience. Does $R \rightarrow \infty$ recover classical mechanics? ^a

Response. No. It only makes the momentum spectrum dense. Classical behavior requires more than closely spaced eigenvalues; in particular, it requires suitable wave packets and a dynamical limit in which position and momentum can both be narrow compared with the relevant macroscopic scales.

^aLecture timestamp: 00:56:58–00:57:20.

Question from the audience. Why does R remain in the momentum spacing but disappear from the angular-momentum spectrum? ^a

Response. A fixed phase winding n around a larger circle has a smaller tangential wave number and hence smaller $p_n = n\hbar/R$. Multiplying by the lever arm R restores $L_n = n\hbar$. The quantized object is the action associated with a full angular circuit.

^aLecture timestamp: 00:57:37–00:58:48.

1.7 Compatibility and Common Eigenstates

A classical particle may be assigned a position and a momentum at the same instant. Quantum mechanically, a position eigenstate is localized like a delta distribution, whereas a momentum eigenstate is a plane wave with uniform position density. These states cannot be the same.

The algebraic question is whether two Hermitian operators $\hat{A}$ and $\hat{B}$ possess a complete common eigenbasis. If

$$\hat{A}|n\rangle = a_n|n\rangle, \quad \hat{B}|n\rangle = b_n|n\rangle, \quad (1.33)$$

then

$$\hat{A}\hat{B}|n\rangle = a_nb_n|n\rangle = \hat{B}\hat{A}|n\rangle. \quad (1.34)$$

Define the commutator

$$[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}. \quad (1.35)$$

Compatible observables commute. Conversely, commuting Hermitian matrices can be simultaneously diagonalized, including within degenerate eigenspaces. For unbounded operators, the same statement requires compatible domains and the corresponding spectral conditions; those analytic details are suppressed here.

1.8 The Canonical Commutator

For position and momentum,

$$(\hat{x}\psi)(x) = x\psi(x), \quad (\hat{p}\psi)(x) = -i\hbar\psi'(x). \quad (1.36)$$

Apply their commutator to an arbitrary differentiable wavefunction:

$$[\hat{x}, \hat{p}]\psi = x\left(-i\hbar\frac{d\psi}{dx}\right) + i\hbar\frac{d}{dx}(x\psi) \quad (1.37)$$

$$= -i\hbar x\psi' + i\hbar(\psi + x\psi') \quad (1.38)$$

$$= i\hbar\psi. \quad (1.39)$$

Since this holds for every state in the common domain,

$$[\hat{x}, \hat{p}] = i\hbar\mathbf{1}. \quad (1.40)$$

The commutator is nonzero on every nonzero state. If a vector were a common eigenvector of $\hat{x}$ and $\hat{p}$, their commutator would annihilate it, contradicting (1.40). Position and momentum therefore have no common eigenvector. The uncertainty relation will later quantify this incompatibility.

Question from the audience. Is every state an eigenvector of the commutator, and can $i\hbar$ be regarded as an observable value? ^a

Response. Every vector is indeed an eigenvector of $[\hat{x}, \hat{p}] = i\hbar\mathbf{1}$, with the same eigenvalue. But the commutator of two Hermitian operators is anti-Hermitian, so it is not itself an observable. The Hermitian combination $-i[\hat{x}, \hat{p}] = \hbar\mathbf{1}$ contains no state-dependent measurement information. Its importance is that it fixes the algebra relating position and momentum.

^aLecture timestamp: 01:15:33–01:16:45.

$$\begin{aligned}
 & i\hbar \left[x \left(-i \frac{\partial}{\partial x} \right) + i \frac{\partial}{\partial x} x \right] \psi(x) \\
 & -i\hbar x \frac{\partial \psi}{\partial x} + i\hbar \frac{\partial}{\partial x} (x \psi(x)) \\
 & i\hbar \psi(x)
 \end{aligned}$$

Figure 1.9: The product rule supplies the uncanceled term in $[\hat{x}, \hat{p}]\psi = i\hbar\psi$.

Lecture frame: 01:13:00.

Question from the audience. Why is $x\psi(x)$ not an eigenvalue times $\psi(x)$?^a

Response. An eigenvalue is one scalar constant for the whole vector. The coordinate x varies across the function. A generic Gaussian, for example, is therefore not an eigenfunction of the position operator. Only a generalized state supported at one point satisfies $x\delta(x - x_0) = x_0\delta(x - x_0)$.

^aLecture timestamp: 01:16:51–01:19:07.

1.9 One Aperture and Two Alternatives

The two-slit experiment turns the linear structure of states into a visible probability pattern. An incoming particle meets a barrier with two small openings, and a screen records the position y at which it arrives. With only one opening available, the outgoing wave spreads. In three dimensions a schematic far-field contribution has the form

$$\psi(r) \sim \frac{e^{ikr}}{r}, \quad r = \sqrt{\ell^2 + y^2}, \quad (1.41)$$

where ℓ is the distance from aperture to screen. Over a small region of the screen it may be approximated by a local plane wave.

Now open both holes. Let $\psi_1(y)$ and $\psi_2(y)$ be the amplitudes associated with the two alternatives. The state-space rule is

$$\psi(y) = \psi_1(y) + \psi_2(y), \quad (1.42)$$

not the addition of the separate probabilities. Locally, suppressing slowly varying envelopes and temporarily setting $\hbar = 1$, write

$$\psi_1 = e^{ipy}, \quad \psi_2 = e^{iqy}. \quad (1.43)$$

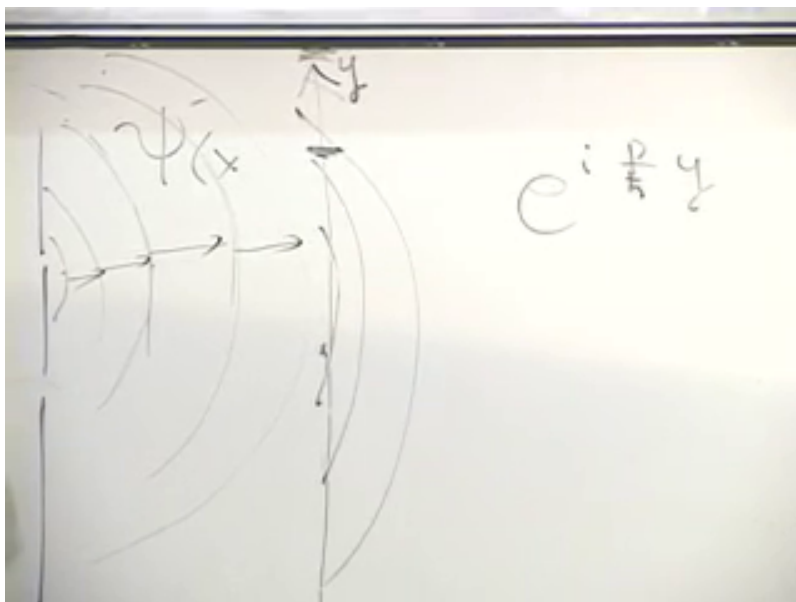


Figure 1.10: A small opening produces an outgoing wave whose local phase gradient changes across the screen.

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Then

$$|\psi_1 + \psi_2|^2 = (e^{ipy} + e^{iqy})(e^{-ipy} + e^{-iqy}) \quad (1.44)$$

$$= 2 + e^{i(p-q)y} + e^{-i(p-q)y} \quad (1.45)$$

$$= 2 + 2 \cos((p - q)y). \quad (1.46)$$

The cross terms retain the relative phase. They create alternating bright and dark fringes even though either local plane wave separately has constant magnitude.

In the far field, if the slit separation is d , the path difference at angle θ is approximately $d \sin \theta$. For wavelength $\lambda = 2\pi/k$, adjacent bright fringes satisfy

$$d \sin \theta_m = m\lambda. \quad (1.47)$$

At small angles, $y \approx \ell \theta$, so

$$\Delta y \approx \frac{\lambda \ell}{d}. \quad (1.48)$$

Moving the holes farther apart makes the fringes closer; moving them together spreads the fringes apart.

Question from the audience. What happens in the local model if $p = q$? ^a

Response. Then the relative phase is constant: $\psi_1 + \psi_2 = 2e^{ipy}$. The intensity is the constant 4, in the chosen normalization. There are no alternating fringes, but the two contributions are everywhere fully constructive.

^aLecture timestamp: 01:32:25–01:33:08.

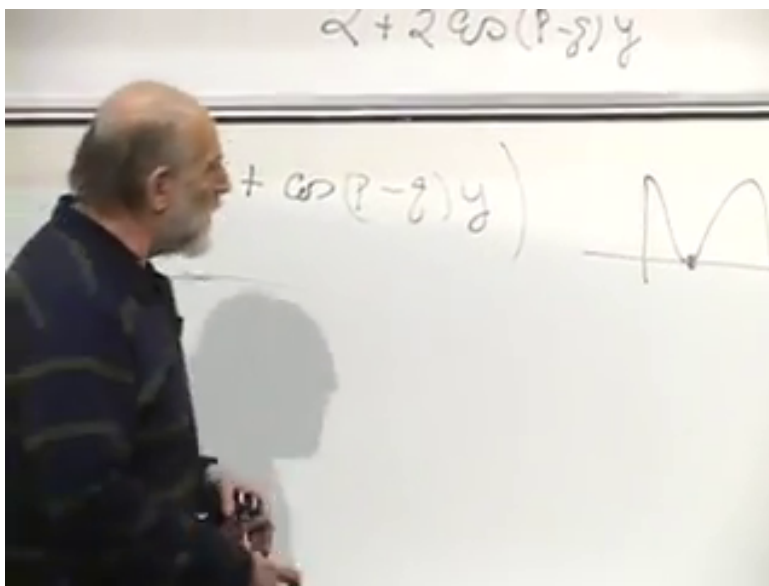


Figure 1.11: The cross terms in the squared sum generate a nonnegative oscillating detection density.

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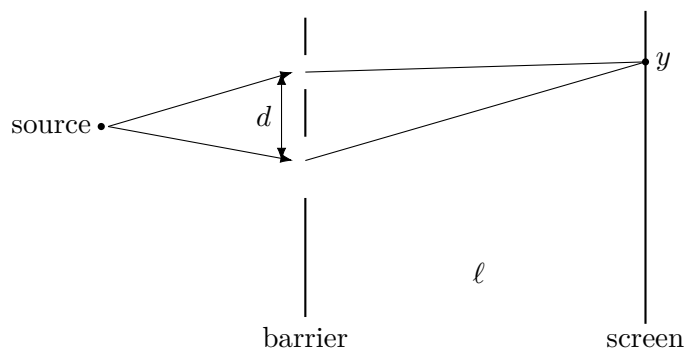


Figure 1.12: The relative path length from two separated openings changes with the screen coordinate.

Question from the audience. Is this the Fraunhofer diffraction pattern? ^a

Response. It is the far-field two-aperture part of that pattern. The local beat model isolates the rapidly varying two-path fringes. A full Fraunhofer calculation also retains the finite width of each opening, which multiplies those fringes by the broader one-aperture diffraction envelope.

^aLecture timestamp: 01:38:39–01:39:57.

1.10 Which-Path Information Must Wait

Suppose a detector at the barrier records which opening the electron used. The detector must then become correlated with the alternative. Schematically,

$$|\Psi\rangle = |\text{path 1}\rangle |D_1\rangle + |\text{path 2}\rangle |D_2\rangle. \quad (1.49)$$

When the detector records are distinguishable, the electron alone no longer has the same interference term. A full explanation requires composite systems and entanglement, so it belongs after the tensor-product structure has been developed.

Question from the audience. Why does a detector placed at the openings destroy the interference pattern?^a

Response. Detecting the path means leaving different imprints in another physical system. The electron and detector must then be described as one composite state, with each path correlated with a different detector record. The interference disappears when those records are distinguishable. The mechanism is entanglement, whose derivation must wait until composite systems are available.

^aLecture timestamp: 01:40:04–01:41:53.

1.11 Phase Gradients and Elastic Deflection

For a wave of the form $e^{i\phi(y)}$, the local vertical momentum is the phase gradient

$$p_y(y) = \hbar \frac{d\phi}{dy}. \quad (1.50)$$

Near the central axis the distance $r = \sqrt{\ell^2 + y^2}$ changes slowly with y , so the vertical phase gradient is small. Farther from the axis, the wave has a larger vertical momentum component. This does not require an increase in kinetic energy. In an elastic scattering picture, the magnitude of the momentum remains fixed:

$$p_x^2 + p_y^2 = p^2. \quad (1.51)$$

An increase in $|p_y|$ is accompanied by a decrease in $|p_x|$; the momentum vector changes direction rather than length.

Question from the audience. Where does the energy come from when the particle acquires vertical momentum?^a

Response. It need not acquire additional energy. The barrier can redirect the momentum elastically, much as a stationary wall changes the direction of a bouncing ball. The outgoing wavelength, and therefore the magnitude of the momentum, can remain the same while its horizontal and vertical components change.

^aLecture timestamp: 01:41:59–01:44:12.

1.12 Particles Arrive One at a Time

The wavefunction describes both alternatives, but each detection event is a localized spot. Send one electron at a time and the screen records one spot at a time. Repeat the preparation many times and the accumulated spots approach the density $|\psi_1 + \psi_2|^2$. The density cannot be negative because it is an absolute square, but it can vanish where the two amplitudes cancel.

Question from the audience. Is it then meaningless to speak of a corpuscular particle in this experiment?^a

Response. No. The particle aspect is visible in each localized detection. The wave aspect is visible in the probability law governing many such detections. A single event is a spot; the interference pattern is the statistical arrangement of many spots.

^aLecture timestamp: 01:44:19–01:45:35.

Question from the audience. How does this compare with the interference of light?^a

Response. The same distinction appears with photons. An intense beam draws the fringe pattern quickly. If the beam is attenuated enough, the screen registers separated photon events, yet those events accumulate into the same interference distribution. Reducing the flux changes the number of particles present at once, not the underlying probability law.

^aLecture timestamp: 01:45:39–01:47:17.

1.13 The One-Aperture Envelope

A single opening does not give a featureless amplitude. A convenient local form is

$$\psi_1(y) = \rho(y)e^{i\phi(y)}, \quad (1.52)$$

where $\rho(y)$ changes slowly while the phase may change rapidly. Its detection density is

$$|\psi_1(y)|^2 = \rho(y)^2. \quad (1.53)$$

The rapid phase cancels against its complex conjugate, leaving a smooth envelope. With a second displaced opening,

$$\psi_2(y) = \rho_2(y)e^{i\phi_2(y)}, \quad (1.54)$$

and the probability becomes

$$|\psi_1 + \psi_2|^2 = \rho_1^2 + \rho_2^2 + 2\rho_1\rho_2 \cos(\phi_1 - \phi_2). \quad (1.55)$$

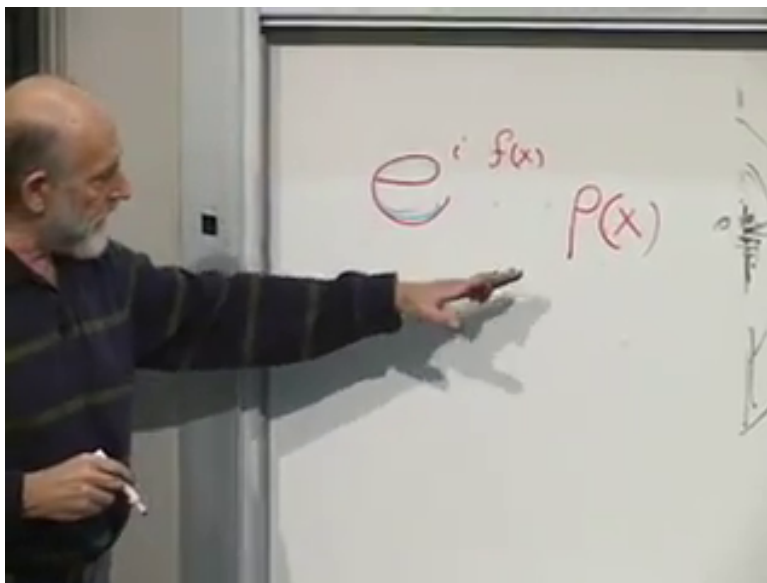


Figure 1.13: A rapidly varying phase multiplied by a smooth envelope gives a smooth one-aperture probability density.

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The smooth one-aperture envelopes are modulated by the relative-phase fringes.

Question from the audience. If the amplitude oscillates behind one opening, why does the screen show only a broad blob?^a

Response. The oscillation is in complex phase. Multiplication by the complex conjugate removes that phase, leaving the slowly varying $\rho(y)^2$. With two openings, a relative phase remains in the cross term, and that is what produces fringes.

^aLecture timestamp: 01:47:28–01:51:36.

A finite aperture can also diffract by itself. Different points across a sharp opening contribute different phases, so a single-slit envelope may contain maxima and minima of its own. This is distinct from the finer two-slit modulation, although a real experiment displays both at once.

Question from the audience. Can the two edges of one sharp opening create their own interference pattern?^a

Response. Yes. A finite opening is a continuum of secondary contributions, and its edges make the phase structure especially visible. The resulting single-aperture diffraction determines the broad envelope; the separation of two openings determines the additional two-path fringes.

^aLecture timestamp: 01:51:37–01:52:41.

1.14 Visibility, Wavelength, and Idealization

Exact dark fringes require coherent contributions of equal magnitude. If the two apertures have unequal amplitudes A_1 and A_2 , then

$$I(y) = |A_1|^2 + |A_2|^2 + 2|A_1 A_2| \cos \Delta\phi(y). \quad (1.56)$$

The minimum is $(|A_1| - |A_2|)^2$, so unequal illumination fills in the dark fringes. Noise, finite source coherence, detector resolution, and imperfect geometry do the same. Under ideal conditions the minima can nevertheless be extremely dark.

Question from the audience. Are there screen positions that will never be hit, even when particles are sent one at a time? ^a

Response. In the ideal equal-amplitude model, yes: destructive interference makes the probability exactly zero at the dark fringes. Real apparatus introduces background, unequal amplitudes, and finite resolution, so observed minima are generally small rather than mathematically perfect.

^aLecture timestamp: 01:52:43–01:53:43.

Question from the audience. Do unequal hole sizes matter, and how small must the openings be? ^a

Response. Small, similarly illuminated openings give the clearest two-path pattern. Unequal sizes alter the two amplitudes and reduce fringe visibility. The relevant scales are the opening widths, their separation, the propagation distance, and the de Broglie wavelength $\lambda = h/p$. Long-wavelength particles diffract through larger structures; short-wavelength electrons require correspondingly finer geometry.

^aLecture timestamp: 01:53:44–01:55:54.

Passing through a sufficiently small opening prepares an approximate outgoing state centered at that opening. In this limited sense the later pattern can become insensitive to many details of the source. Coherence still matters: the source and apparatus must not carry distinguishable which-path information.

Question from the audience. Does the particle forget its source after passing through the small opening? ^a

Response. The opening can prepare a new approximate outgoing state, so many earlier details cease to affect the downstream shape. That is an idealization, not complete independence from the source: momentum spread, coherence, and correlations with other systems can still matter.

^aLecture timestamp: 01:55:55–01:56:41.

1.15 Reuse Is Not Reversal

The final question separates two ideas that are easy to confuse. One may collect an electron, prepare it again, and repeat the experiment with that same particle; repeated trials do not require newly created electrons. Reversing an entire detected event is a different matter. The screen and all other correlated degrees of freedom would have to be included in the reversal.

Question from the audience. Could the same electron be sent through repeatedly, and is that the same as reversing the experiment? ^a

Response. The same electron may be recovered, re-prepared, and used for another trial, producing the same statistics as any identically prepared electron. Reversing a completed detection is not the same operation. To decide whether the full evolution can be reversed, one must first introduce Hamiltonians, time evolution, and the detector as part of the quantum system.

^aLecture timestamp: 01:56:42–01:59:16.

That is where the calculation properly stops. The kinematics has supplied continuous bases, delta normalization, quantized momentum on a circle, noncommuting position and momentum, and interference from added amplitudes. The next step is dynamics: a Hamiltonian must say how these states evolve in time.